Does weather impact Citibike usage in NYC?

Avishai Narsey

DATAENGINE CASE STUDY

The questions

For Citibike operations, who decide fleet size, rebalancing and maintenance.

01 How much does the season drive demand?

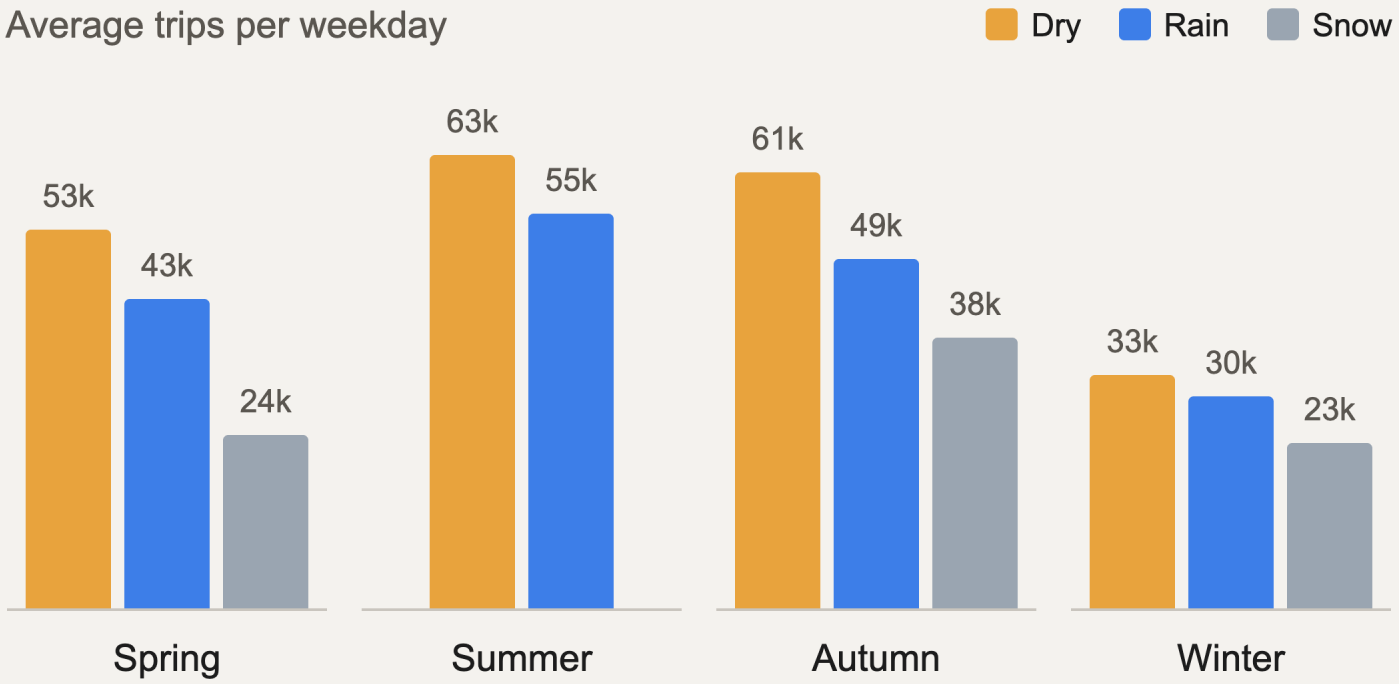
02 How much do rain and snow cut trips on top of that?

03 Does it matter how long it rains?

04 Which riders change their behaviour?

05 What should operations do about it?

Season first, weather second



~2×

trips on a summer weekday vs a winter weekday

~15%

fewer trips on rain days, same season and day type

~40%

fewer trips on snow days, same season and day type

Architecture

Amazon S3
376 trip files (CSV), 61 weather files (JSON)

→

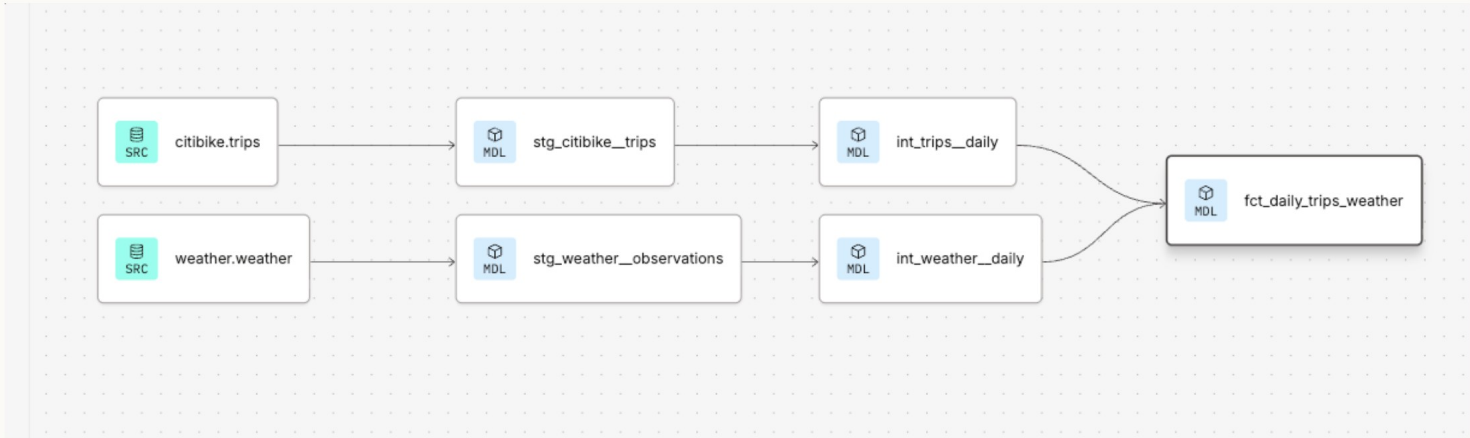
Snowflake RAW
Loaded as-is with COPY INTO

→

dbt Cloud
Staging, intermediate, mart

→

Streamlit
Running inside Snowflake



What the data needed

Found	Handled
Weather mixed NYC with a point in upstate New York	Kept NYC only, by city ID
27% of NYC weather readings were duplicates	Kept one copy of each
Temperature in Kelvin, time in UTC	Converted to °C and New York time
19 trips logged twice by the docks	One trip per bike per start time
5 days with no weather readings	Left out

Modelling decisions

Three layers

Staging cleans, intermediate reshapes, the mart serves the dashboard.

One row per day

The grain both sources share. 721 days in the final table.

Rain from the label

Two-thirds of rain readings had no amount recorded.

Trips as a table

61 million rows. As a view, every read redid the work.

Testing and deployment

PULL REQUEST

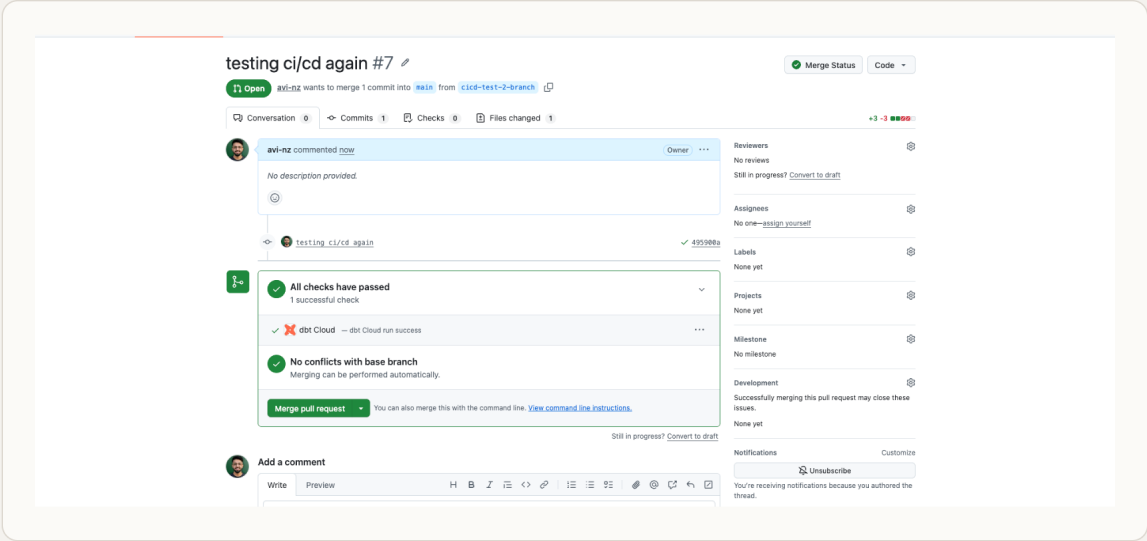
CI builds and tests the changed models

MERGE TO MAIN

Production rebuilds what changed and everything downstream

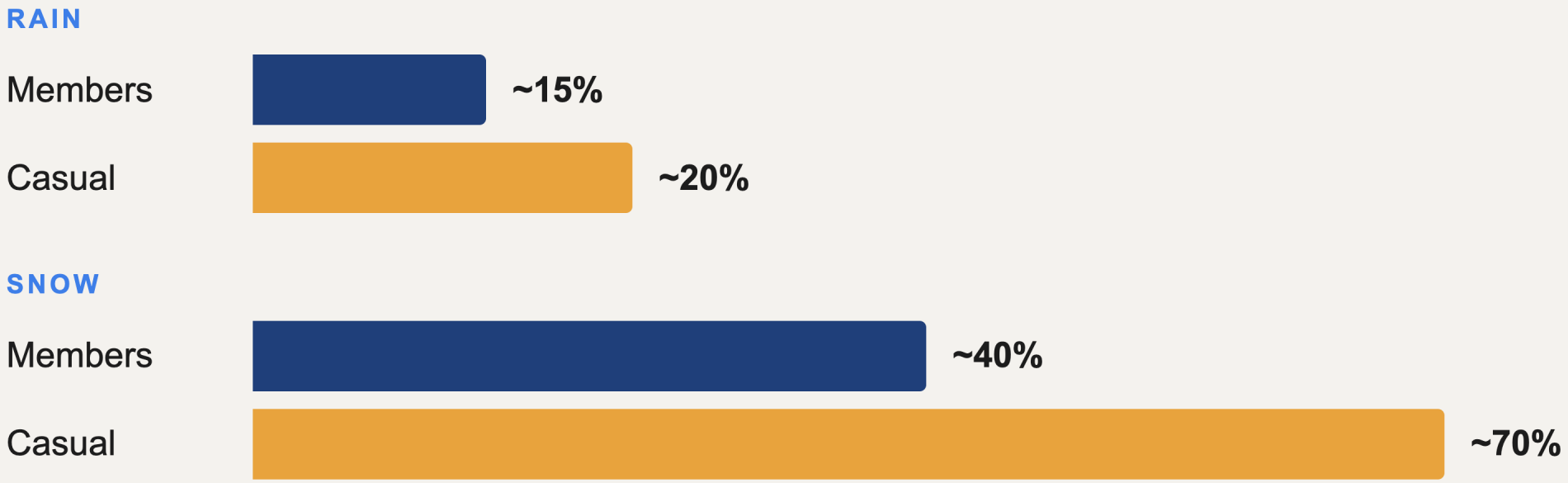
TESTS

A failing unique test is how I found the duplicate weather readings



Casual riders feel the weather most

Fewer trips than on dry days in the same season and day type.



Live demo

The seasons · Weather on top · Who is affected · What to do

Assumptions and limits

- Correlation, not cause
- Holidays and events not controlled for
- "Season" mixes temperature, daylight and tourism
- "Dry" means no rain or snow, so it includes cloudy days
- Only July 2016 to June 2018, where both sources overlap

Next steps

ANALYSIS

- Hourly model for rain at commute times
- Public holidays as a dbt seed
- Regression separating temperature, rain and day type
- Newer Citibike data with matching weather

PLATFORM

- Auto-deploy the dashboard on merge
- Separate CI environment and service user
- Read-only reporting role for the dashboard

Thank you

Questions

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